

EncoderX Remote Internship

Batch 02 | Track: AI/ML | Week 01

Task 1: Image Classification Model

Intern Name: Hazem Emam

Date: September 2026

Project Overview

An end-to-end Computer Vision classification pipeline built with PyTorch on the Intel Image Classification dataset (6 classes: *buildings*, *forest*, *glacier*, *mountain*, *sea*, *street*). The model features a custom Convolutional Neural Network (CNN) with 3 convolutional blocks, max-pooling, dropout regularization ($p = 0.5$), and fully connected classification layers.

Key Performance Metrics

- **Dataset Size:** 16,900 images (70% Train, 15% Validation, 15% Test)
- **Overall Accuracy:** 86% on the 2,535 test images
- **Macro Average F1-Score:** 0.86
- **Top Class:** Forest (Precision: 0.97, Recall: 0.94, F1-Score: 0.96)

Project Deliverables & Links

- **GitHub Repository Link:**
<https://github.com/Hazem-Emam-404/EncoderX-Internship/tree/main/Assignment-1>
- **Project Demonstration Video Link:**
<https://drive.google.com/file/d/1mIfto7e7yqQz1JJltveIDWR02hqV9Nfw/view?usp=sharing>
- **LinkedIn Post Link:**
<https://www.linkedin.com/in/hazem-emam-645436272/>
- **Trained Model File Link:**
<https://drive.google.com/file/d/1LJHCCxP2-V0hoPzPou3XF0hKajJTshNy/view?usp=sharing>

Submitted in partial fulfillment of the EncoderX AI/ML Remote Internship requirements.